

Topic 2.9 - Logarithmic Expressions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Evaluate logarithmic expressions by translating between logarithmic and exponential forms.
- Approximate common, natural, and arbitrary-base logarithms with technology.
- Interpret logarithms as exponents in mathematical, scientific, and contextual scales.

Section A: Foundation and representation

Question 1

Core skill**No calculator**

Rewrite each statement in exponential form: (a) $\log_2 32 = 5$; (b) $\ln e^{-3} = -3$; (c) $\log_{1/3} 9 = -2$.

Question 2

Core skill**No calculator**

Rewrite each statement in logarithmic form: (a) $7^2 = 49$; (b) $10^{-4} = 0.0001$; (c) $e^{1/2} = \sqrt{e}$.

Question 3

Core skill

No calculator

Evaluate exactly: $\log_5 625$, $\log_8 2$, and $\log_{1/4} 64$.

Question 4

Core skill

No calculator

Evaluate $\log_{16}(1/8)$ exactly.

Question 5

Core skill

No calculator

Solve $\log_b 64 = 2$ for the positive base b .

Question 6

Core skill

No calculator

Solve $\log_3(x - 4) = 5$.

Section B: Application, comparison, and error analysis

Question 7

Core skill

No calculator

State the domain of $\log_2(x^2 - 9)$.

Question 8

Core skill

Calculator permitted

Approximate $\log_5 73$ to four decimal places using change of base.

Question 9

Reasoning

No calculator

Order $\log_2 12$, $\log_3 12$, and $\log_{10} 12$ from least to greatest without calculating decimals.

Question 10

Context

No calculator

The decibel model is $D = 10 \log(I/I_0)$. How many decibels correspond to intensity $1000I_0$?

Question 11

Error analysis

No calculator

A student claims $\log_2(-8) = -3$ because $2^{-3} = 1/8$. Diagnose the error.

Question 12

Scale interpretation

Calculator permitted

A base-10 logarithmic scale increases from 2.4 to 5.1. By what factor does the underlying quantity increase?

Section C: AP-style multiple choice**MCQ 1**

No calculator

What is $\log_4(1/64)$?

- A. -3
- B. -2
- C. 2
- D. 3

MCQ 2

No calculator

Which is equivalent to $\log_7 x = 3$?

- A. $x^7 = 3$
- B. $7^x = 3$
- C. $7^3 = x$
- D. $3^7 = x$

MCQ 3

No calculator

Which expression has domain $x > 5$?

- A. $\log(x - 5)$
- B. $\log(5 - x)$
- C. $\log(x^2 - 25)$
- D. $\log(x + 5)$

MCQ 4

No calculator

If $\log_b 20 = 2.5$, then:

- A. $b^{2.5} = 20$
- B. $20^{2.5} = b$
- C. $b^{20} = 2.5$
- D. $2.5^b = 20$

Section D: Free-response reasoning**FRQ 1 - Exact and approximate logarithms**

Calculator permitted

6 points

(a) Evaluate $\log_{27} 9$ exactly. (b) Solve $\log_4(x + 3) = -2$. (c) Approximate $\log_6 145$ to three decimal places. (d) State the domain of $\ln(2x^2 - 18)$.

FRQ 2 - Logarithmic scale

Calculator permitted

6 points

The model $S = 20 \log(A/A_0)$ compares an amplitude A with reference amplitude A_0 . (a) Find S when $A = 100A_0$. (b) Solve for A/A_0 when $S = 54$. (c) Compare the underlying amplitude ratios for scale values 30 and 50. (d) Explain why adding 20 to the scale multiplies rather than adds to the amplitude ratio.
